

VR EDUCATION PROGRAM: DETAILED STORYBOARDS (M1–M3)

Project: SIMPLE (NDICI ASIA/2022/438-619) Pedagogical Model: Discovery-Based Learning (Failure-Consequence-Success) Technical Framework: GAMA Simulation Backend / Unity Frontend Total Program Duration: 240 Minutes (4 Hours)

Module 1: The Mosquito Survival Challenge (Urgency)

Goal: Internalize the biological drive of the *Aedes* mosquito and the natural/human threats to its offspring.

Phase 1: The Adult Stage (Aerial Gameplay)

- **Scene 1: The Awakening (Tutorial):** Player spawns as a female *Aedes*. Focus: Learn flight controls.
- **Scene 2: Seeking a Mate:** Follow audio cues to find a male.
- **Scene 3: The Mating Dance:** Match flight frequency. **Success:** Abdomen pulses; status: "Fertilized."

Phase 2: The Hunter (Stealth & Feeding)

- **Scene 4: The Hunger:** Track CO2 trails (green smoke) to find a human host in the garden.
- **Scene 5: The Blood Meal (Human Predator):** Land and suck blood.
 - **The Danger Meter:** Fills based on human movement and shadows.
 - **The Attack:** If the meter peaks, the human NPC will slap or use a racquet. **If hit, the player dies.**
 - **Relay Trigger:** Death immediately triggers the **Team Relay Mechanic** (Headset switch).
- **Scene 6: Heavy Flight:** Fly engorged (slower speed) to find a water container. Avoid mid-air predators (dragonflies).
- **Scene 7: The Cycle Completed (Egg Laying):**
 - **Action:** Player must choose a water source: **Large Jar (Hidden)**, **Old Tire**, **Small Pot (Visible)**, or **Natural Pond**.
 - **Mechanic:** Tap to lay eggs.
 - **Information UI:** *"Cycle complete! These eggs will hatch in 7–10 days. Beware: Humans and predators may destroy them!"*
 - **Relay Trigger:** Successful egg-laying triggers the **Team Relay Mechanic**.

GAMA: Survival Calculation

- **The Result Screen:** Displays "Fate of your Eggs" (Predator/Human randomized events).
- **The Mosquito Explosion:** Final survivors are sent to the GAMA model for Module 2. **GAMA Hook:** The number of surviving eggs becomes the initial mosquito density for Module 2.

Inter-Module Transition: From Vector to Protector

- **Visual:** The "Big Racket" zap effect. Screen flashes white.
- **Animation:** 7-day time-lapse.
- **Role Shift:** Player becomes the **Action Detective**. HUD changes to the "Action Scanner."

Module 2: The Aedes Action Detective (Prevention)

Goal: Apply the "5 Por" (5P) measures and utilize personal protection to survive a mosquito swarm.

- **Scene 1: The Inspection:** Search the yard for breeding sites.
- **Scene 2: The 5 Por Measures:** Use Pid (Cover), Plian (Change), Ploi (Release), Pumbung (Improve), Patibat (Practice).
- **Scene 3: The Safe Zone (House Interior):**
 - **Recharge Mechanic:** Resting inside restores "Energy."
 - **Protective Gear:** Find and equip long-sleeved clothing.
 - **Environmental Control:** Close windows, turn on AC, and use Fans to create "No-Fly Zones."
- **Scene 4: The Defense (The Breaking Point):**
 - **Dynamic Swarm:** Population scales based on M1 survivors and M2 prevention thoroughness.
 - **Relay Trigger:** Reaching 10 bites (X amount) triggers an immediate **Team Relay Mechanic**.

GAMA: Cleared Breeding Sites

- The final **Source Reduction %** (breeding sites cleared) is sent to the GAMA model to calculate the disease transmission rate for Module 3.

Inter-Module Transition: The Community Briefing

- **Visual:** Map of Vientiane village hotspots.
- **Narrative:** "The mosquitoes have spread. We need you to check on your neighbors."

- **GAMA Hook Integration:** The density and location of the red "Hotspots" are a direct result of the containers missed in Module 2.

Module 3: The Vientiane Health Leader (Community Outreach)

Goal: Proactively manage a community health crisis by managing NPC referrals and identifying dangerous self-medication.

- **Scene 1: The Neighborhood Walk:** A 3D Vientiane village street.
 - **GAMA Data Hook:** The number of sick NPCs (lethargic/rubbing heads) is generated by GAMA based on the **Source Reduction %** from M2. Low cleanup = High caseload.
- **Scene 2: The Health Scan (Proactive Triage):** Walk up to neighbors and use the "Scanner" to evaluate their health.
 - **Decision Matrix:** Based on symptoms (High fever + stomach pain/vomiting), the player must choose:
 - **Clinic Referral (Red Gate):** For urgent cases showing "Alarm Signals."
 - **Home Care (Green Gate):** For mild cases. Instruction: "Rest, Drink ORS, use a Bed Net."
- **Scene 3: The Pharmacy Gatekeeper (Medication Safety):** Intercept NPCs with Red Icons (Aspirin/Ibuprofen). Physically swap meds for Paracetamol.
- **Scene 4: High-Risk Identification (Secondary Infection):** Identify neighbors who are at extreme risk of DHF due to medical history.

Technical & UI Requirements

- **HUD: * M1: Hunger/Danger Bars.**
 - **M2: Energy Bar (Health) and Bite Counter (X/10).**
 - **M3: Neighborhood Safety Index (%) and Referral Accuracy Counter.**
- **Team Relay Mechanic (2:1 Ratio):** Triggered by death, 10 bites, cycle completion, or 5-minute timer.
- **GAMA Inter-Module Logic (M2 to M3):**
 - $\text{Outbreak_Intensity} = (100 - \text{M2_Source_Reduction_Percent}) * \text{Population_Density_Factor}$.
 - If $\text{M2_Source_Reduction} < 50\%$, the number of "Critical" triage cases in M3 doubles.

Certified by: Benjamin Barutzki CEO, BeDev Consulting Sole Co., Ltd. **Revision Date:** January 27, 2026